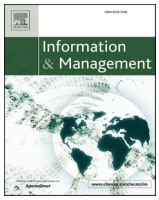
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Momentum effects in crowdsourcing contest platforms

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ABSTRACT

Members of a crowdsourcing contest platform usually participate in multiple contests over time. Their performance in these contests can create a performance trajectory, producing positive or negative momentum. Although such momentum is a pervasive phenomenon in practice, it has largely been overlooked in the crowdsourcing contest literature. The broader literature on momentum posits that it can affect individuals' cognition and change their behavior. We leverage insights from this literature to investigate how momentum influences contestants' performance and the underlying mechanisms. We show that contestants' momentum is positively associated with their contest performance by motivating contestants to engage in more learning activities (including learning from others and learning from own experience) between contests and exert more effort during contests. By adopting momentum theory as a novel theoretical lens, our study enriches the crowdsourcing contest literature by showing that momentum is a novel and impactful determinant of individuals' contestant performance. Besides, whereas classical momentum theory asserts that motivation explains the positive relationship between momentum and performance, we extend the boundary of momentum theory by showing that contestants' motivation manifests in their learning activities and effort in the context of crowdsourcing contest platforms. Taken together, our study offers important practical implications for the design and governance of crowdsourcing contest platforms.

1. Introduction

Organizations are increasingly using online crowdsourcing contest platforms, such as Kaggle, 99designs, and Innocentive, to seek solutions for a variety of problems [1,2]. These platforms organize contests for organizations to tap into the wisdom of the crowd to solve their problems [3,4]. As crowdsourcing contests are a pertinent form of organizations' open innovation strategy [5,6], a sizable body of literature has investigated the various factors that could facilitate or impede individuals' participation and performance in crowdsourcing contest platforms [1,7,8]. These factors can be broadly categorized into two groups: attributes of the participants (e.g., motivation, social capital) and design parameters of the contests (e.g., contest duration, contest prize, provision of feedback).

Notably, a significant research gap remains in this stream of literature. That is, prior studies capture the previously mentioned factors at a given point of time or a period but seldom consider the intertemporal *change* of these factors [1]. This perspective is important and relevant in crowdsourcing contests, because it is a prevalent phenomenon that members of a crowdsourcing contest platform participate in multiple contests over time, producing a series of records [9–11]. The most pertinent and noticeable record is contestants' performance scores [12–14]. Although prior studies have touched upon some aspects of contestants' past performance records by investigating how contestants' past performance, past success, and past failure impact their subsequent performance or behavior, these constructs mainly reflect contestants' *levels* of quality, ability, or experience [10,15]. What are ignored is the *change* (increase or decrease) of past performance scores and the resultant psychological state of mind on contestants [16,17]. The change in contestants' performance trajectory matters because it can generate psychological momentum (henceforth, *momentum*), which refers to a sense of psychological power that can shape a person's perception of herself [18]. The psychological power associated with momentum can influence contestants' subsequent performance because psychological momentum can greatly alter one's motivation [16,17,19,20].

Building on insights from momentum theory, we hypothesize that individuals' momentum can strengthen contestants' motivation, thus elevating their performance in contests [17,21]. Moreover, to unpack

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motivation in the context of crowdsourcing contests, we further propose that contestants' motivation can manifest in their learning activities (including learning from others and learning from own experience) [14,22] between contests and effort during contests [23,24], which are two context-specific mechanisms that explain the positive relationship between momentum and contest performance. Based on our analyses of archival data collected from Kaggle, we find support for all the propositions.

Our theoretical contributions are twofold. First, by adopting momentum theory as a novel theoretical lens, our study enriches literature on crowdsourcing contest by showing that momentum engendered by the change of past performance records is an impactful determinant of contestants' performance [3,4,13,14,22]). Besides, our research draws researchers' attention to explore how the change of some factors can impact contestants' performance. Second, our study also extends the boundary of momentum theory [16–18]. Prior studies on momentum are mainly conducted in contexts such as sports and laboratory [16,25–27]. Although this stream of research has shown that motivation mediates the momentum-performance relationship, motivation can be reflected in different activities in different settings. By applying momentum theory in the context of crowdsourcing contest platforms, we uncover that contestants' motivation manifests in their learning activities and effort between or during contests, which are two context-specific mechanisms underlying momentum-performance relationship. These two context-specific mechanisms contribute to a more nuanced understanding of the mechanisms of momentum. Taken together, our study offers important implications for the design and governance of crowdsourcing platforms.

2. Theoretical background

Our study builds on two realms of research: crowdsourcing contests and momentum.

2.1. Crowdsourcing contests

In his seminal work, Howe [28] used crowdsourcing to refer to an instance of a company making an open call to outsource a task that was once performed by its employees to a large number of individuals outside the company. A common form of crowdsourcing is crowdsourcing contests [10,13,29,30]. Crowdsourcing contests are mostly organized by online platforms, such as Kaggle and 99designs [31]. A typical process of a crowdsourcing contest is as follows [3,23]. First, a company identifies a specific and challenging problem or task that the company considers can benefit from the wisdom of the crowd. Second, the company communicates with a crowdsourcing contest platform and discusses the design (e.g., description of task, prize, dataset, duration) of a crowdsourcing contest. Third, the crowdsourcing contest platform launches the contest and broadcasts the contest to attract a large number of participants. Finally, the contest winner transfers the winning solution (e.g., machine learning algorithm, logo design) to the company when the contest is completed. In a nutshell, crowdsourcing contests enable companies to leverage the wisdom of the crowd to generate innovative solutions [7]. Crowdsourcing contests are particularly suitable for problems that are highly demanding on technical, analytical, or design skills [32]. The crowdsourcing contest platform is an intermediary that connects the company (i.e. solution seeker) and the contestants (i.e. solution provider) [10,33].

Given that crowdsourcing contests are an important open innovation approach for firms to seek external innovative ideas and solutions [30], research has investigated various factors that can impact contestants' participation and performance [34]. These factors are related to design parameters of the contests and the characteristics of the contest participants.

One vein of research has investigated the effects of contest design parameters on contestants' participation and performance. These design parameters include contest recommender system [4], solution exemplar [35], prize guarantee [36], crowd voting [9], problem specification [29], contest duration [10], provision of feedback [29], introduction of online platform [33], task instruction writing strategies [11], online reputation system [37], contest prize [38], competitiveness [39], linguistic signals in task description [40], the usage of AI [41], seeker-generated information [42], and so on. These studies covered the general design parameters of a contest (e.g., prize, duration), features of the online platform that the contest is hosted on (e.g., reputation system, online knowledge community), contest organizers' organizing strategies (e.g., feedback giving, task description), and emerging technologies (e.g., the incorporation of AI).

Another stream of research has investigated the impact of contestant participants' characteristics on their participation and performance in crowdsourcing contests. These characteristics include motivation [30], behavioral contribution patterns [43], Machiavellianism [44], alignment of social and intellectual capital [13], salience bias [3], strategic effort allocation [45], self-efficacy [23], problem-seeker knowledge distance [46], competition with superstars [47], cognitive load [12], learning from others [14], social dominance-based faultlines [15], status [48], social value orientation [49], liability of foreignness [50], lost time [51], and so forth. Overall, these studies have examined a range of social, behavioral, and psychological factors as well as contest strategies employed by contestants.

In Table 1, we present a summary of representative articles that explored different factors that influence contestants' participation or performance in crowdsourcing contests. Our focus on contestants' momentum is directly linked to the second stream of literature that look at contestants' attributes. Momentum is a novel factor in this line of research because it captures the change of contestants' track records (i.e., performance record in our study) [18,52], whereas the factors in extant literature are captured at a point of time or a period of time (e.g., by taking average). In the subsection that follows, we review the literature on momentum to delineate what it is and how it can pollinate the literature on crowdsourcing contests.

2.2. Momentum

Momentum (psychological momentum) refers to a sense of psychological power that can shape a person's perception of herself as well as what will happen in the future [18]. It is a felt state of mind in which an individual perceives that things will move unstoppably toward a direction following the previous trajectory [20]. Accordingly, momentum is bidirectional [17]. Whereas a series of successful events can form a positive trajectory, which engenders positive momentum, a series of unsuccessful events can produce a negative trajectory, which induces negative momentum [20]. We focus on momentum induced by past performance (i.e., performance momentum) in the present study, which we define as individuals' sense of increased or decreased psychological power in response to their gain or loss of performance in the previous period of time [52,53]. Even though performance momentum and past performance, past success, or past failure capture aspects of performance in the previous period of time, they are different constructs for three reasons. First, performance momentum is induced by the positive or negative *change* of performance in the performance trajectory formed by performance records in the previous period [16,25], whereas past performance, past success, or past failure reflects average quality of performance based on these performance records [10,15]. For example, performance records of 30[t-2]−60[t-1]−90[t] and 90[t-2]−60[t-1]−30[t] at three different points of time indicate positive and negative performance separately, whereas they signify similar level of quality or ability because the mean values are 60 in both cases. Second, whereas performance momentum is a subjective construct that reflects individuals' perception or state of mind [21], past performance, past success, or past failure is an objective construct that represents individuals' competence or incompetence [54–56]. Third, performance

Table 1
Representative articles that investigated the factors impacting crowdsourcing contest participation or performance.

Category	Article	Independent variable (IV)	Dependent variable (DV)	Effect of IV on DV
Contest design parameters	Chen et al. [9]	Crowd voting	Contest participation	Positive
	Chen et al. [10]	(1) Contest duration	Contest participation	(1) Positive
		(2) Contest prize		(2) Positive
	Jian et al. [36]	(1) Prize guarantees	Contest participation	(1) Positive
		(2) In-process feedback		(2) Positive
	Koh & Cheung [35]	Seeker exemplars	(1) Contest performance	(1) Negative
			(2) Contest participation	(2) Negative
	Mo et al. [4]	Recommender system	Performance	Positive
	Yan & Hollingshead [49]	Reward structure	Contestant performance	Positive/negative
	Ye & Jensen [33]	Introducing an online community	Contestant performance	Positive
Contestants' attributes	Cao et al. [15]	Social dominance-based faultlines	Contest participation	Positive/negative
	Cheng et al. [12]	(1) Germane cognitive load	Contest performance	(1) Positive
		(2) Intrinsic cognitive load		(2) Negative
		(3) Extraneous cognitive load		(3) Negative
	Dissanayake et al. [13]	(1) Leader intellectual capital	Contest performance	(1) Not significant
		(2) Leader social capital		(2) Positive
		(3) Expert intellectual capital		(3) Positive
		(4) Expert social capital		(4) Not significant
		(5) Alignment between social capital and intellectual capital		(5) Negative
	Hutter et al. [44]	Machiavellianism	Contest participation	Positive/negative
	Jin et al. [14]	(1) Volume of shared knowledge	Contest performance	(1) Negative
		(2) Quality of shared knowledge		(2) Positive
		(3) Generativity of shared knowledge		(3) Positive
	Lee et al. [3]	Salience bias	Contest performance	Negative
	Pollok et al. [46]	Problem-seeker knowledge distance	Contest participation	Inverted U-shaped
	Zhang et al. [47]	Competing with a superstar	Contest performance	Positive
	Zheng et al. [30]	(1) Participation intention	Contest participation	(1) Positive
		(2) Intrinsic motivation		(2) Positive

momentum impacts individuals' projection of what will happen in the future [52]. Whereas positive performance momentum empowers the individuals to project that such positive trajectory will move toward a desirable direction, negative momentum engenders a perception that such downward trend will continue in the next period of time. In comparison, past performance, past success, or past failure merely indicates individuals' current level of competence or skills [15,23]. Table 2 summarizes these conceptual differences.

Momentum has long been studied in psychology [19,25,57], where researchers have shown that momentum leads to various outcomes related to performance, cognition, and behavior [34]. First, some studies explored the relationship between a focal individual's momentum and her performance. The general finding in this literature is that momentum enhances performance. For example, Nahm et al. [58] found that salespeople's momentum elevates their subsequent performance. In a different setting, Zhao and Zhang [59] found that archers' momentum is positively related to their subsequent performance. Moving from the perspective of a focal individual to an observer's perspective, Kakkar et al. [53] presented evidence that a focal individual's positive momentum poses a threat to her opponent, which detracts from the opponent's performance. Apart from performance, momentum has also been shown to relate to cognition and behavior. For example, in an experimental study, Pettit et al. [52] found that holding constant current status, individuals with a positive status momentum are granted higher status by observers than those with a negative momentum.

Table 2
Conceptual differences between performance momentum and past performance and past success/failure.

	Performance momentum	Past performance/past success/past failure
Relevance with past performance	Performance change	Average performance level
Subjective/Objective construct	Subjective (state of mind)	Objective
Relevance with future outcomes	Direct projection or perception of future outcomes	Current level of competence or skills

3. Theory and hypotheses

In this section, we first elucidate the rationale behind adopting the momentum perspective to study performance and the choice of learning and effort as the two mediating mechanisms in the context of crowdsourcing platform. Then we theorize the relationship between momentum and performance as well as the mediating effects of learning and effort in this relationship.

3.1. Research model

We adopt momentum theory to study contestants' performance as momentum is a pervasive phenomenon in crowdsourcing contest communities [11,60]. Specifically, as contestants enter multiple contests sequentially at different points of time, their performance scores in these contests can form a trajectory, creating a scenario for the contestants to infer their momentum [52,61]. Whereas an upward trajectory induces perception of positive momentum, a downward trajectory engenders perception of negative momentum [20]. Here we make an assumption that contestants can perceive their momentum on the basis of the past performance trajectory. This assumption is justified by research in social psychology, which shows that individuals can accurately perceive momentum based a series of consecutive records. For instance, Vallerand et al. [21] designed a scenario study that shows participants a list of performance scores of tennis players. They found that participants in this study can infer tennis players' momentum based on their performance trajectory. Likewise, Pettit et al. [52] found in several controlled experiments that individuals perceive positive momentum based on an ascending trajectory of ranks (e.g., 8–7–6) and perceive negative momentum based on a descending trajectory of ranks (e.g., 4–5–6).

Classical momentum theory holds that momentum can enhance individuals' performance by elevating their motivation [18,21,27]. Motivation refers to individuals' inner desire to make an effort or to take action [62]. Motivation is a general psychological mechanism that can be reflected by concrete activities in a specific context. For instance, research has shown that motivation can manifest in different activities in the contexts of school [63], sports [64], workplace [65], and marketing

[66]. On the basis of this line of research, our focus on the context of crowdsourcing contest communities creates an opportunity to extend the boundary of momentum theory by showing its mechanisms in this specific context [17]. Drawing on literature on crowdsourcing contests, we contend that motivation can manifest in two types of concrete activities in the crowdsourcing contest communities: contestants' learning activities between contests and their effort during contests [23,24]. We focus on these two types of activities for the following two reasons.

First, we consider that learning activities can reflect contestants' motivation to perform well in crowdsourcing contests [14]. As crowdsourcing contest is a knowledge intensive setting where contestants leverage their expertise to formulate creative solutions to solve contest organizers' problems [10]. To sustain their competitive advantages in contests, it is key for contestants to keep learning or keep acquiring new knowledge so as to keep their knowledge base up to date [47]. For learning activities, we draw insights from recent IS literature to categorize them into two groups: learning from others and learning from own experience [14,47]. Learning from others can be further split into three types based on recent IS literature, namely quantity of learning sources, quality of learning sources, and generativity of learning sources [14]. In line with traditional experiential learning theory, learning from own experience refers to learning by practicing in crowdsourcing contest communities [67].

Second, contestants' effort reflects another important aspect of motivation [23]. This is because crowdsourcing contests usually last for months, during which contestants can expend effort to experiment with multiple solutions in order to identify an optimal one [68]. Contestants who have a high level of motivation should make effort to try more solutions [30]. On the contrary, those who have a low level of motivation should be less prone to invest effort in this trial-and-error process [23].

In summary, based on our survey of literature on momentum and literature on crowdsourcing contest, we propose that the performance enhancing effect of momentum is explained by their motivation in two types of activities in crowdsourcing contest: learning activities between contests and effort during contests. Our theoretical model is depicted in Fig. 1.

In the rest of this section, we will theorize the main effect of momentum on contest performance as well as the two types of mechanisms in the context of crowdsourcing contest platform.

3.2. The main effect of momentum on performance

Even though a positive relationship between momentum and performance might seem apparent, findings in the momentum literature that speak to it are mostly based on small-scale datasets or laboratory experiments [17,21], warranting a test in a new empirical context such as the crowdsourcing contest platform we focus on. In addition, in this new context, we can also explore the mechanisms underlying the momentum-performance relationship, both to expand the boundary of research on momentum and also to understand how momentum might influence performance in this important setting.

In a crowdsourcing contest community, contestants can enter multiple contests over time [23]. Their performance records are closely monitored by these contestants. At a given point, contestants know their performance records in the few precedent contests they have completed. These performance records depict a trajectory for contestants to infer their current momentum. For instance, performance trajectory 80–50–20 indicates positive momentum, whereas performance trajectory 10–40–70 denotes negative momentum. It is not clear whether contestants can intuitively perceive their current momentum based on past performance records. Research in social psychology has shown direct evidence that inference of momentum based on the outcomes of a few consecutive events is feasible. For example, Vallerand et al. [21] manipulated momentum by creating hypothetical scenarios. In the positive momentum scenario, a player loses in the first few games and then wins in the remaining games. In the no-momentum scenario, a player wins a game and then loses a game and this pattern repeats. The manipulation checks in this research suggest that individuals can accurately perceive their momentum on the basis of a trajectory of outcomes in past events.

We propose that momentum has a performance-enhancing effect. This happens because people tend to project future outcomes based on past results [69]. For contestants with positive momentum (i.e. an ascending performance trajectory), they expect that this positive movement of their performance will not stop, and they expect their performance to continue to increase. Such perception engenders elevated psychological power, which enhances motivation [17,27,70]. We postulate that the motivation induced by momentum can manifest in several concrete activities in a crowdsourcing contest platform. These activities help contestants maintain their performance or perform better in the next contest [27,61]. In crowdsourcing contests, contestants can strategically engage in two types of activities in response to their momentum. First, momentum can drive contestants to invest more time in learning activities. Learning is an effective strategy to enrich the knowledge base of contestants and keep their skills up to date [47]. For this reason, contestants with momentum will be motivated to spend more time searching for relevant knowledge, identifying valuable knowledge, and assimilating it. However, as knowledge acquisition takes a considerable amount of time and effort to search, screen, and understand the relevant knowledge, momentum can provide the psychological force for contestants to carry through this process [71]. Second, momentum can also stimulate contestants to put in more effort during a focal contest, because this is likely to directly enhance their

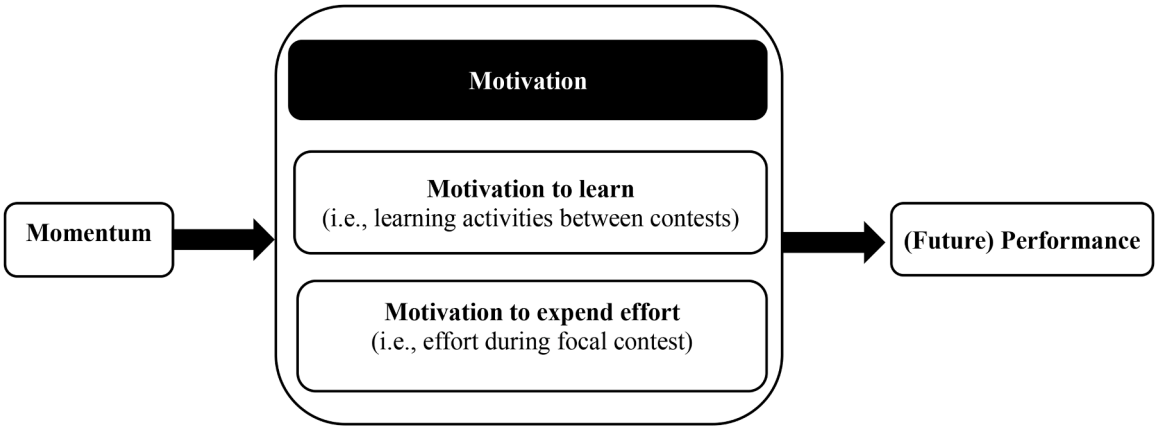


Fig. 1. Research model.

performance. In crowdsourcing contests, effort can be reflected in the number of contestants' trials during a contest [45]. Arguably, as the number of trials increases, the likelihood for them to identify an optimal solution can increase, since contestants can learn through this experimentation process [72]. Based on these arguments, we propose that:

Hypothesis 1. Contestants' momentum is positively related to their future contest performance.

3.3. The mediating effect of learning activities

Having proposed the main effect of momentum on contest performance, we now theorize the mediators underlying this relationship. Based on our discussion at the start of the hypotheses section, we contend that the positive association between momentum the performance is mediated by contestants' motivation, which manifest in two types of concrete activities: learning activities between contests and effort during focal contest.

Momentum can motivate contestants to engage in activities in two periods that we call the interval period and the contest period. As shown in Fig. 2, the interval period is from the end of the previous contest to the start of a focal contest. The contest period refers to the period between the start of the focal contest until the end of that focal contest. The learning period consists of both the interval period and the contest period, such that it covers the period from the end of the previous contest to the end of a focal contest. We consider the end of the previous contest as a starting point for the learning period, because that is the time when the contestants are made aware of their performance in that previous contest. Accordingly, they can then make an inference about their momentum based on that performance and their performance in previous contests. This learning period continues until the end of the focal contest, because that is the time when the contestants' momentum can be updated based on their performance in the focal contest.

During the learning period, momentum motivates contestants to leverage the various resources that are available in the crowdsourcing contest community to strengthen their knowledge base. Specifically, they can "learn by viewing" and "learn by practicing" using the resources available in the crowdsourcing contest community [22]. Contestants "learn by viewing" by browsing the technical resources (e.g., code scripts) shared by others, identifying those that they deem useful, going through the technical resources, and absorbing the knowledge contained in such resources. To "learn by viewing," contestants need to invest time and effort to search, identify, and absorb knowledge. We argue that contestants with greater (positive) momentum have the psychological power to engage in this process to strengthen their skills, such that they can maintain or enhance their performance in subsequent contests.

The effectiveness of "learning by viewing" can vary depending on the characteristics of the sources that the learners view. Prior research suggests that the learning sources that contestants view can be characterized along three dimensions: knowledge quantity, knowledge quality, and knowledge generativity. Knowledge quantity is the number of learning sources viewed by contestants, knowledge quality refers to the average level of positive evaluation that these learning sources received, and knowledge generativity is reflected by the average number of derivatives generated based on these learning sources [14]. We contend that momentum motivates contestants to seek and learn from a larger amount of resources shared by others, and also from resources that are of higher quality and generativity, because these activities would strengthen their knowledge base of such contestants, thereby contributing positively to their performance in the next contest [73,74].

First, momentum can drive contestants to view more resources that is shared by others. Viewing more resources can help contestants enrich and solidify their knowledge base, subsequently elevating their performance [75]. Typically, a learning resource focused on a specific topic in a knowledge domain. Therefore, the more content that contestants go through, the more likely it is that they can accumulate knowledge in different domains, which increases their knowledge diversity and benefits future performance.

Hypothesis 2a. The positive relationship between contestants' momentum and their future performance is positively mediated by the quantity of learning sources that they view during the learning period.

Second, momentum can stimulate contestants to seek higher-quality learning sources. Higher quality learning sources are more likely to contain knowledge that can assist contestants to acquire deep and valuable insights in certain knowledge domains [76]. In addition, as high-quality learning sources tend to be novel, credible, and relevant, they are conductive to improve performance [77]. Such high-quality learning sources help contestants gain in-depth insights that can then improve their future performance.

Hypothesis 2b. The positive relationship between contestants' momentum and their future performance is positively mediated by the quality of learning sources that they view during the learning period.

Third, momentum can facilitate learning by viewing resources that are higher in generativity. High-generativity resources can aid performance for two reasons. These resources are more adaptable, such that the learner can use them to generate derivatives or produce other solutions based on them [78]. High-generativity resources can also be more easily integrated with the learner's existing knowledge base, facilitating the creation of new knowledge via recombination [14]. In this manner, viewing high-generativity scripts can help contestants absorb knowledge that can be adapted to their existing knowledge base, which strengthens their future performance.

Hypothesis 2c. The positive relationship between contestants' momentum and their future performance is positively mediated by the

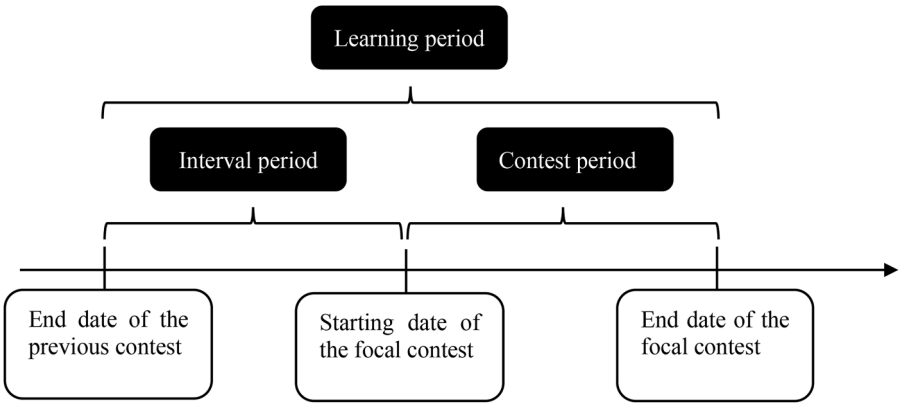


Fig. 2. Illustration of learning period, interval period, and contest period.

generativity of learning sources that they view during the learning period.

Apart from "learning by viewing," momentum can also drive contestants to "learn by practicing." This can happen because crowdsourcing contest communities allow contestants to practice using prior completed contests. Practicing using these contests can help because it provides information about the real contest scenario, where contestants can submit their solutions and receive performance scores or feedback from the system. Based on that feedback, contestants can fine tune their solutions and try again. This iterative process can get contestants to reflect on the strengths and weakness of their solutions. In this way, it promotes searching for solutions to improve the quality of their solutions, leading them to accumulate hands-on skills [79,80]. In short, momentum can also encourage contestants to practice using previous completed contests, which sharpens their skills and contributes to their performance in the next contest.

Hypothesis 2d. The positive relationship between contestants' momentum and their future performance is positively mediated by their learning from practicing during the learning period.

3.4. The mediating effect of effort

During the contest period, momentum can motivate contestants to exert more effort in order to maintain or increase their performance in the focal contest. By effort, we mean the number of solutions that contestants submit during a contest. Crowdsourcing contests usually last for a few weeks or months, thereby contestants can submit multiple solutions during these contests. The number of solutions during a contest matter, because contestants can receive performance feedback for each of their solutions. Based on the performance feedback, contestants can then fine tune their solutions. Specifically, after receiving the performance feedback of a submitted solution, a contestant can evaluate the strength and weakness of their current solution, look for ways to enhance the current solution, implement any such methods in a new solution, and finally submit this new solution to see the performance score. If their performance is now higher, the contestant can continue to fine tune the solution following a similar logic. If their performance drops, the contestant can look for alternative methods to improve their solution. All else being equal, as the number of their trials increases, contestants are more likely to identify a method that helps them to formulate a solution with better performance. In summary, we expect that the positive relationship between contestants' momentum and performance will be explained in part by their effort during the contest.

Hypothesis 3. The positive relationship between contestants' momentum and their future performance is positively mediated by their effort during the contest period.

4. Data and methods

4.1. Research context: kaggle

Launched in April 2010, Kaggle is an online community for individuals who are interested in machine learning techniques. Individuals can engage in two types of activities in this community. First, they can participate in crowdsourcing contests. The problems in these contests are defined by organizations, including major ones such as Google, the National Football League, and American Express. These organizations provide datasets for contestants to train and test their machine learning models. Contestants' performance in contests is a function of the predictive power of their models. In most contests, the top three to five participants can win monetary rewards. On average, a contest lasts for about three months and attracts about 900 participants. Second, in addition to contests, Kaggle also establishes online communities to facilitate individuals' sharing of learning resources (i.e., code scripts) and mutual learning. Specifically, Kaggle has an online community that allows individuals to share their code scripts and view others' code scripts.

4.2. Data

For our empirical analyses, we collected publicly available data from Kaggle (https://www.kaggle.com/kaggle/meta-kaggle). In particular, we have archived data from Kaggle beginning with April 2010, when Kaggle was launched, until July 2022. Since our goal is to examine the impact of momentum on contestants' performance and behavior, we include both solo contestants, i.e., those who entered contests alone, as well as teams, with two or more individuals, in our estimation sample. It is worth noting that Kaggle refers to each separate entrant as a "team," regardless of whether the contestant is a solo entrant or consists of more than one member. Because our calculation of a Kaggle member's momentum is based on her performance in the three contests before the focal contest, we include only those solo entrants who have completed at least three contests before the focal contest and only those teams in which all members completed at least three contests before the focal contest. Based on these considerations, we end up with 106,770 contestant/team-contest observations in our estimation sample, where 103,410 (96.9 %) observations are solos. Here "contestant" refers to solo participants, whereas team refers to team participants (two or more participants), where all team members' data are aggregated by taking average, such that each team is considered as one observation in our sample.

4.3. Dependent variable

Contest performance. We measure *Contest performance* using rank on the private leaderboard in a given contest. Rank has been used in research on crowdsourcing contests to measure contestants' relative performance (e.g., [3,13]). At the end of each contest there is a rank list based on contestants' final performance. Rank varies from 1 (for the best performing contestant) to the total number of contestants that have entered a given contest. Because the total number of contestants varies across contests, we normalize the rank of each contestant to 0–1 range using a min-max normalization. After this normalization, the smaller a contestant's normalized rank, the better their performance. Then, for easier interpretation, we reverse code this normalized rank by subtracting it from 1 and then multiplying it by 100. As a result, contest performance ranges from 0 to 100, where the larger the number, the better the performance. The way we normalize the rank and calculate the contest performance is shown in formula (1) below.

$$\textit{Contest performance}_{ic} = \left(1 - \frac{rank_{ic} - \min\,(rank_c)}{\max\,(rank_c) - \min\,(rank_c)}\right) * 100 \qquad (1)$$

where $rank_{ic}$ is focal contestant i's rank in a contest c. $\min\,(rank_c)$ and $\max\,(rank_c)$are the minimum and maximum rank, respectively, in contest c across all contestants. For solo participants (96.9 % observations), contest performance refers to each individual's own performance (i.e., rank) in a contest. For team participants, Kaggle shows the performance (i.e., rank) achieved collectively by all team members in each team; we therefore use this performance rank to measure the team's contest performance. As noted before, each team is considered as an independent entrant, thus it appears only once in our sample. In other words, we do not consider each team member separately in our sample.

4.4. Independent variables

Momentum. We follow prior studies to measure a contestant's momentum using her performance in the three contests before the focal contest [52,53]. This measure subtracts a contestant's average performance in the three prior contests from her performance in the previous

contest and then divides this by the contestant's average performance in the prior three contests. The formula to calculate momentum is as follows:

$$Momentum_{it} = \frac{P_{it-1} - Avg.\left(\sum_{t-1}^{t-3} P_{it}\right)}{Avg.\left(\sum_{t-1}^{t-3} P_{it}\right)} \quad (2)$$

where P_{it-1} is the ith contestant's performance in the previous contest and $Avg.\left(\sum_{t-1}^{t-3} P_{it}\right)$ is the ith contestant's average rank in the previous three contests. For instance, if a contestant's performance in the previous three contests is 30, 60, and 90. Her momentum is 0.5 ((90 – (30+60+90)/3)/((30+60+90)/3)), indicating positive momentum. In comparison, the momentum for a performance trajectory of 90, 60, and 30 would be −0.5, showing negative momentum.

4.5. Mediating variables

Our mediation model includes four mediators, as detailed next.

Learning. The first four mediators are related to contestants' learning by viewing others' shared code scripts or learning by practicing using previous, already completed, contests. We follow Jin et al. [14] to use three variables to measure contestants' learning from others' shared code scripts. The first variable captures the quantity of learning sources, which is measured by the number of scripts that the focal contestant viewed between the end of the previous contest that she entered and the end of the current contest. The second variable is the quality of learning sources, which we measure using the average number of upvotes of the scripts that a focal contestant viewed between the end of the previous contest that she entered and the end of the current contest. The third variable measures the generativity of learning sources, which takes the average number of forks of the scripts that a focal contestant read between the end of the previous contest that she entered and the end of the current contest. The fourth variable captures a contestant's learning from their own experience, which is operationalized by the number of solutions submitted by a contestant in completed contests between the end of the previous contest that she entered and the end of the current contest. These four variables are log-transformed due to their skewed distributions.

Effort. Our fifth mediator, effort is measured by the number of submissions made by a contestant during the focal contest. This variable is log-transformed because of its skewed distribution.

4.6. Control variables

Team size. This variable is measured by number of members in the teams.

Past performance. We average the contestant's performance (i.e., normalized ranks on the private leaderboard) in all the contests that she entered before the focal contest to measure her past performance.

Experience. We measure experience as number of contests that a contestant entered before the focal contest. We log-transformed this variable due to its skewed distribution.

Skill. We follow the literature to measure contestants' skill using their skill points that are calculated by Kaggle based on their achievements in past contests [45].[1] Specifically, the point earned by a contestant in a contest is shown in formula (3):

$$\left[\frac{100000}{\sqrt{N_{teammates}}}\right]\left[Rank^{-0.75}\right]\left[log_{10}\left(1 + log_{10}\left(N_{participants}\right)\right)\right] \quad (3)$$

where $N_{teammates}$ refers to number of team members in a team (it is equal to 1 for solo participants). *Rank* represents a solo contestant's or a team's performance rank in a contest. $N_{participants}$ refers to number of participants in a contest, including both teams (each team is counted once) and solo participants. Besides, Kaggle implemented a decay function for the points, which is shown below. The t represents number of days elapsed since the point was awarded.

$$e^{-t/500} \quad (4)$$

Skill is operationalized by the total points (multiplied by the decay function) earned by a contestant in all the contests that she entered before focal contest.

Status. Kaggle has a formal hierarchy system in which members obtain tiers based on their achievements in past contests. The hierarchy system has five hierarchical levels (from low to high): *Novice, Contributor, Expert, Master*, and *Grandmaster*, which we capture with a measure that ranges from 0 to 4.

Contribution of UGC at t-1. This variable refers to lagged contribution of technical content, interactive content, or data content, which is operationalized by contestant's contribution of UGC in the previous period of time.

4.7. Method

To test Hypotheses 1–3, we employ *PROCESS* macro in *R* developed by Hayes [81], which is widely used in IS research to test mediation effects (e.g., [82,83]). This method is suitable to test our mediation model that has multiple parallel mediators [81]. Specifically, we enter all mediators simultaneously in the same model to test the mediating effects.

5. Results

5.1. Descriptive statistics and correlations

We present the descriptive statistics and correlations in Table 3. We examined the variance inflation factors (VIFs) to assess multicollinearity. The average VIF is 1.64 and the maximum VIF is 3.55, suggesting that multicollinearity is not a significant concern in our analysis.

5.2. Effect of momentum on performance (Hypothesis 1)

In Hypotheses 1–3, we proposed that momentum is positively associated with performance and that this relationship is mediated by six factors. First, we investigated the main effect of momentum on performance using *PROCESS* macro in *R*. We entered the dependent variable, independent variable, and control variables, but did not include the mediators. This analysis showed that momentum has a positive and significant effect on performance ($\beta = 2.015$, $p < 0.001$), supporting Hypothesis 1.

5.3. Mediating effects for the momentum-performance relationship (Hypotheses 2–3)

To test the mediation effects in Hypotheses 2 and 3, we conducted a parallel mediation analysis by including all six mediators in the same model. The results are shown in Fig. 3. First, the quantity of learning sources positively mediates the relationship between momentum and performance ($\beta = 0.035$, $p < 0.001$), supporting Hypothesis 2a. In terms of the strength of this effect, this mediator accounts for 4.4 % of the total mediating effect. Second, the quality of learning sources positively mediates the relationship between momentum and performance ($\beta = 0.043$, $p < 0.001$), supporting Hypothesis 2b This mediator accounts for 5.5 % of the total mediating effect. Third, the generativity of learning sources positively mediates the relationship between momentum and performance ($\beta = 0.009$, $p < 0.001$), supporting Hypothesis 2c, and

[1] https://www.kaggle.com/progression

Table 3
Descriptive statistics and correlations ($N = 106{,}770$).

		Mean	S.D.	1	2	3	4	5	6	7	8	9	10	11	12	13
1	Performance	57.8	28.16													
2	Momentum	0	0.41	0.03												
3	Quantity of learning sources	1.12	1.33	0.14	0.03											
4	Quality of learning sources	2.62	2.47	0.12	0.03	0.74										
5	Generativity of learning sources	0.72	1.01	0.08	0.01	0.45	0.69									
6	Learning from experience	0.43	0.91	0.12	0.03	0.18	0.11	0.09								
7	Salience bias in focal contest	0.07	1.42	−0.01	0.01	0.00	−0.01	−0.01	0.00							
8	Effort in focal contest	1.95	1.43	0.52	0.02	0.13	0.10	0.07	0.13	0.13						
9	Team indicator	0.03	0.17	0.16	0.01	0.08	0.08	0.03	0.09	0.01	0.22					
10	Past performance	60.35	17.84	0.41	−0.01	0.16	0.14	0.06	0.16	0.02	0.24	0.12				
11	Experience	2.37	0.84	0.11	0.01	0.17	0.10	0.05	0.10	0.03	0.02	0.06	0.22			
12	Specialization	0.06	0.12	0.02	0.00	−0.07	−0.06	−0.03	−0.02	0.07	0.05	−0.01	−0.03	−0.09		
13	Skill	8.15	1.99	0.24	0.01	0.04	−0.03	0.02	0.15	0.03	0.15	0.11	0.40	0.55	−0.09	
14	Status	1.27	0.99	0.21	0.00	0.15	0.13	0.04	0.14	0.01	0.14	0.19	0.48	0.57	−0.07	0.50

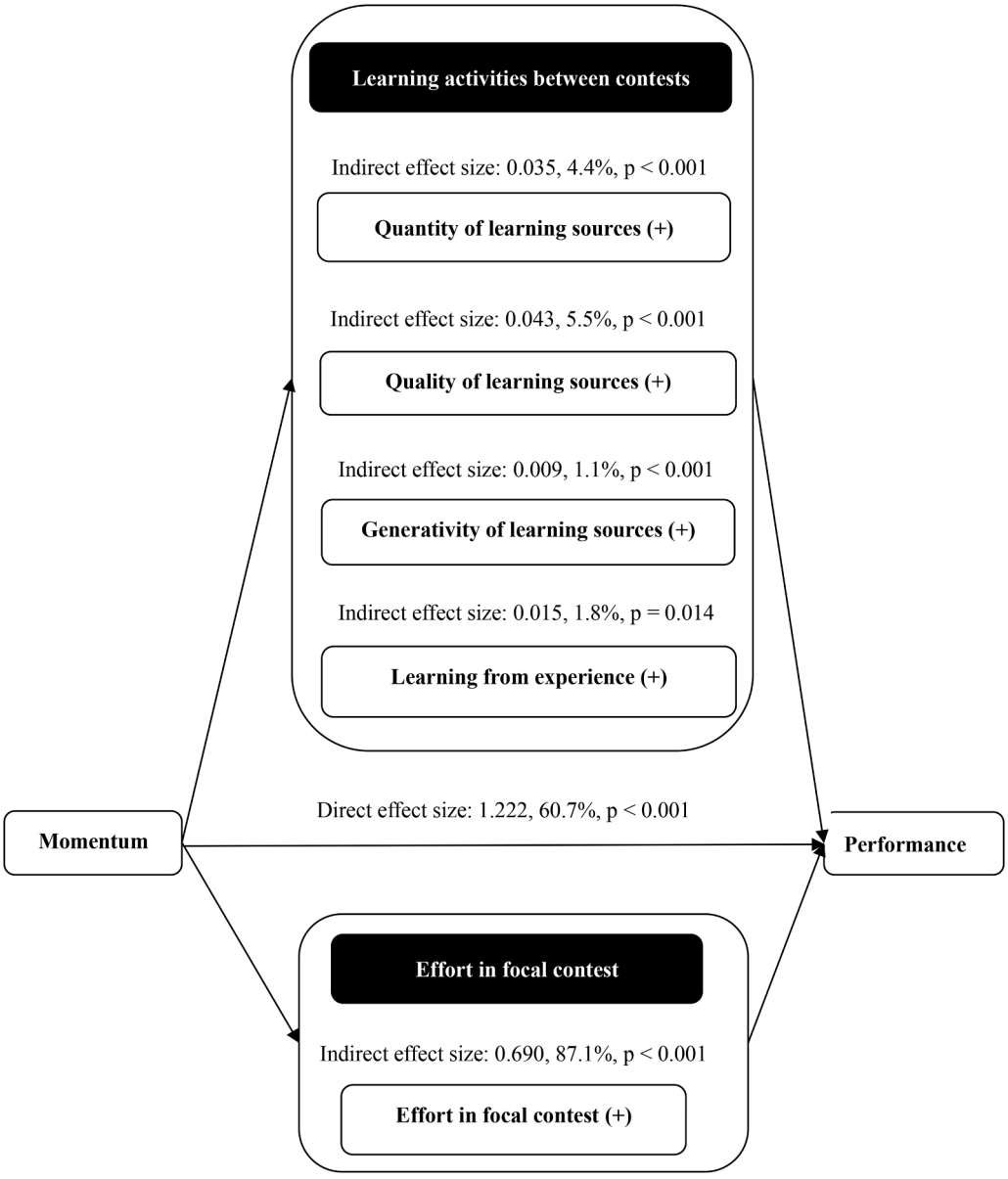


Fig. 3. Mediation analyses.

accounting for 1.1 % of the total mediating effect. Fourth, learning from experience positively mediates the relationship between momentum and performance ($\beta = 0.015$, $p < 0.001$), supporting Hypothesis 2d, and accounting for 1.8 % of the total mediating effect. Overall, all four factors that are related to learning positively mediate the momentum-performance relationship, and account for 12.9 % of the mediating effect in total.

Fifth, effort in focal contest positively mediates the relationship between momentum and performance ($\beta = 0.690$, $p < 0.001$), supporting Hypothesis 3. This is the strongest mediator, accounting for 87.1 % of the total mediating effect.

5.4. Robustness checks

We conducted three sets of checks to assess the robustness of our findings.

First, the estimation sample in our main analyses included both solos and teams. Since a great majority (96.9 %) of the contest participants are solo contestants, we reran all the models on a sample that included solo contestants only. The results are shown in Fig. A1 of the appendix document, with all the hypotheses remain supported ($p < 0.001$ for H1, H2a, H2b, and H3; $p = 0.014$ for H2c; $p = 0.003$ for H2d).

Second, in our main analyses, we operationalized momentum based on the previous three contests. As two alternatives to this number, we recalculated momentum using the previous two and previous four contests. The results are shown in Figs. A2-A3 of the appendix document. All the hypotheses continue to receive support if momentum is measured using the previous two contests ($p < 0.001$ for H1, H2a, H2b, and H3; $p = 0.046$ for H2c; $p = 0.002$ for H2d). All the hypotheses, except for H2d, remain supported if momentum is measured using the previous four contests ($p < 0.001$ for H1, H2a, and H3; $p = 0.014$ for H2b; $p = 0.016$ for H2c; $p = 0.085$ for H2d).

Third, we operationalized contest performance by normalizing contestants' performance ranks in the contests. An alternative measure of contest performance is contestants' performance scores on the Kaggle leaderboard. It is worth noting that the evaluation metrics to calculate these performance scores vary across contests. For some evaluation metrics (e.g., accuracy, R-squared, area under the curve, Matthews correlation coefficient), higher score indicates better performance, whereas for the other evaluation metrics (e.g., root mean squared error, log loss, mean absolute error), lower score indicates better performance. We implement the following two steps to normalize the performance scores to ensure that higher score consistently indicates better performance. For contests that aims to maximize performance scores, we normalize contestants' scores using formula (5). For contests that aim to minimize performance scores, we transform contestants' scores using formula (6).

$$\textit{Contest score performance}_{ic} = \left(\frac{score_{ic} - \min(score_c)}{\max(score_c) - \min(score_c)}\right) * 100 \quad (5)$$

$$\textit{Contest score performance}_{ic} = \left(1 - \frac{score_{ic} - \min(score_c)}{\max(score_c) - \min(score_c)}\right) * 100 \quad (6)$$

where $score_{ic}$ is focal contestant i's score in a contest c. $\min(score_c)$ and $\max(score_c)$are the minimum and maximum score, respectively, in contest c across all contestants. We reran the analyses for H1-H3 based on this alternative performance measure. We present the results in Fig. A4 of the appendix document. The results continue to lend support to the H1-H3 ($p < 0.001$).

6. Discussion and implication

Drawing from momentum theory, we postulate that momentum produced by contestants' change of past performance is an impactful determinant of their contest performance. Using archival data from Kaggle, the current research shows that a contestant's momentum can enhance their performance by motivating them to engage in more learning activities (learning from others and learning from own practicing or experience) between contests and exert more effort during contests.

Our mediation analyses reveal a few interesting findings. First, the mediation effect of effort is considerably stronger than the mediation effect of learning. Second, among the learning activities, learning from others has a stronger mediation effect than learning from experience. Third, among the activities of learning from others, quantity and quality of learning sources has stronger mediation effects than generativity of learning sources. Finally, we performed three groups of robustness checks, including restricting sample size, varying measurement time window, and trying alternative dependent variable. All these analyses confirm that our findings are robust.

6.1. Theoretical contributions

The current research makes two main theoretical contributions. First, we adopt momentum theory as a novel theoretical lens to study its performance implication in the context of crowdsourcing contest communities. To the best of our knowledge, our study is the first one that employs this theory in the literature on crowdsourcing contest [1]. Momentum is a psychological force that is perceived by individuals based on a series of objective events that happened in the past [27]. Therefore, momentum is a less explicit factor compared to other factors that have been investigated in prior literature, such as contest prize [10], contestants' social capital [13], and contestants' motivation [30]. This novel theoretical perspective deepens our understanding that momentum engendered by the change of past performance records is an impactful determinant of contestants' performance [3,4,13,14,22]. Besides, as momentum theory is a classical perspective in social psychology [21], it can draw researchers' attention to the many theories in social psychology in the future, which broadens the theoretical scope to investigate various phenomena in crowdsourcing contest communities. Finally, our study also draws researchers' attention to explore how the change in contestants' track records can impact their psychology and behavior, leading to different outcomes [52,53].

Second, we uncover that learning and effort are two types of novel mechanisms underlying the momentum-performance relationship. In line with the classical momentum theory, we posit that momentum can increase individuals' performance by enhancing their motivation [18,21,27], which manifests in two types of concrete activities in crowdsourcing contest communities: learning activities between contests and effort during contests [23,24]. These mechanisms suggest that momentum is a type of psychological force that drives individuals to engage in knowledge acquisition activities (i.e., learning) and trial-and-error activities (i.e., effort). These two groups of mechanisms echo with the extant views in the literature on crowdsourcing contests. The first group of mechanism (i.e., learning activities) is consistent with the current view that crowdsourcing contest is a competitive and knowledge intensive context, where involvement in active learning is a core determinant of success [14]. The second group of mechanism (i.e., effort) is in alignment with the perspective that crowdsourcing is a strategy of open innovation, where a trial-and-error process facilitates the identification of better solution [84]. These context-specific mechanisms enrich our understanding of the mechanisms of momentum in the context of crowdsourcing context communities, which extends the boundary of momentum theory [17].

6.2. Practical implications

Our findings offer important and concrete practical implications for different stakeholders on crowdsourcing contest platforms in the current digital and AI era [85], which include the platform, the contest

organizers (solution seekers), and contestants (solution providers). In the rest of this section, we will discuss the practical implications for these three groups of stakeholders separately. First, from the perspective of the platform, as individuals' contest performance is positively related to their momentum, the crowdsourcing contest platform could utilize information about momentum strategically. For contestants with positive momentum, platforms can prominently remind them of this momentum (e.g., via notifications or emails) to encourage them to perform better, while for contestants with little or negative momentum, platforms can make the information about their momentum less salient to them to prevent their momentum from reducing their motivation in contests.

Second, for contest organizers, since we find that contestants' effort during contest is a strong mediator for the positive momentum-performance relationship, the contestant organizers can take various strategies to motivate contest participants to exert more effort during contests. For example, the contest organizers can view contestant participants' submitted solutions and provide customized feedback regarding how they can further optimize their solutions. This can provide directions for contestants formulate new solutions. Besides, the contest organizers can also share other relevant materials, such as academic papers and exemplar code scripts, such that contestant participant can refer to these materials to formulate more solutions. Finally, since the amount of reward in contests is set by the contest organizers, a straightforward and effective approach to stimulate more effort from participants is to provide higher amount of monetary incentive.

Third, from the perspective of contestants, since learning from knowledge shared by others and learning by practicing are two of the mechanisms that underlie the momentum-performance relationship, contestants can actively view knowledge (especially knowledge sources with high quality and high generativity) shared by other users and practice using past contests, so as to keep their knowledge base up to date. For contestants with no or negative momentum in particular, they are prone to be demotivated in learning activities. However, our findings imply that momentum itself does not differentiate contest performance between contestants, instead, it is contestants' learning and effort that determine their final performance in contests. Therefore, even though some contestants do not possess desirable momentum, they can also achieve decent performance by engaging in more learning activities and expend more effort during contests.

6.3. Limitations and future research directions

Our study is not without limitations. First, we tested our hypotheses in only one crowdsourcing contest platform. Future research can explore the generalizability of our findings in other crowdsourcing contest platforms. Second, we focus on how the change of performance impacts contestants' subsequent performance in the present study. Future research can explore the implications of the change of other relevant factors, such as the change of feedback, network relationships, and status [86]. Third, we followed prior studies (e.g., [53]) to look at how individuals' momentum, understood in a recent sense, influences their performance and behavior. It is possible that an individuals' momentum can change over a longer course of time. Future research can investigate whether patterns of change of momentum over a longer stretch of time impacts performance and behavior. Fourth, while we uncovered that learning and effort bias mediate the momentum-performance relationship, future research can examine other mediators underlying this relationship.

7. Conclusion

To conclude, we drew insights from the momentum literature to investigate how the change of contestants' performance impacts their subsequent performance in a crowdsourcing contest platform. We contribute to the crowdsourcing literature by showing that momentum is a pervasive and important factor that affects contestants' performance. We also enrich momentum theory by showing its context-specific mechanisms on crowdsourcing contest platforms. Our study also offers practical implications for crowdsourcing platforms to engage their users based on their momentum.

CRediT authorship contribution statement

Tengjian Zou: Writing – review & editing, Writing – original draft, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Zike Cao:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix Document

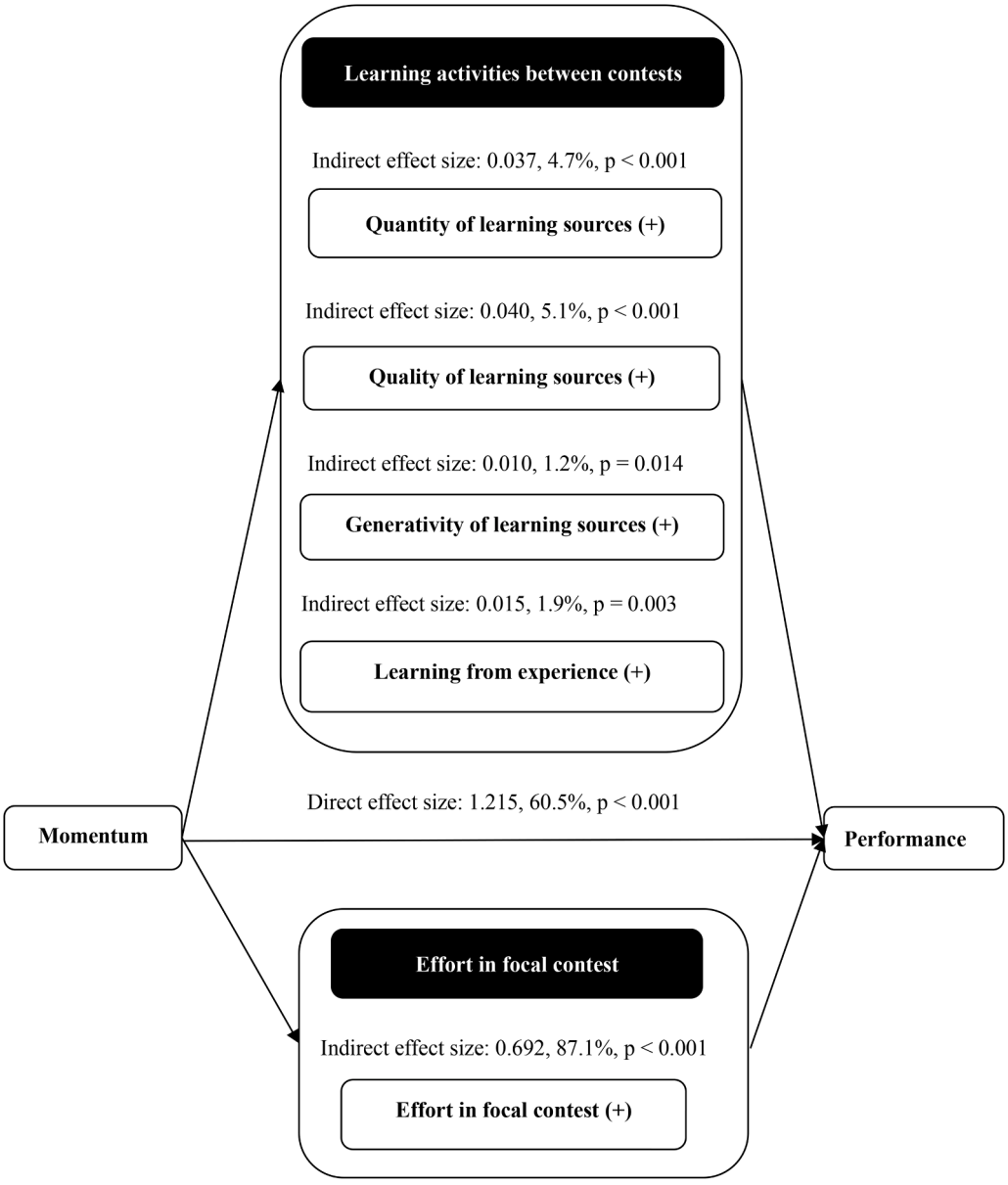


Fig. A1. Mediation analyses (sample is restricted to solo participants).

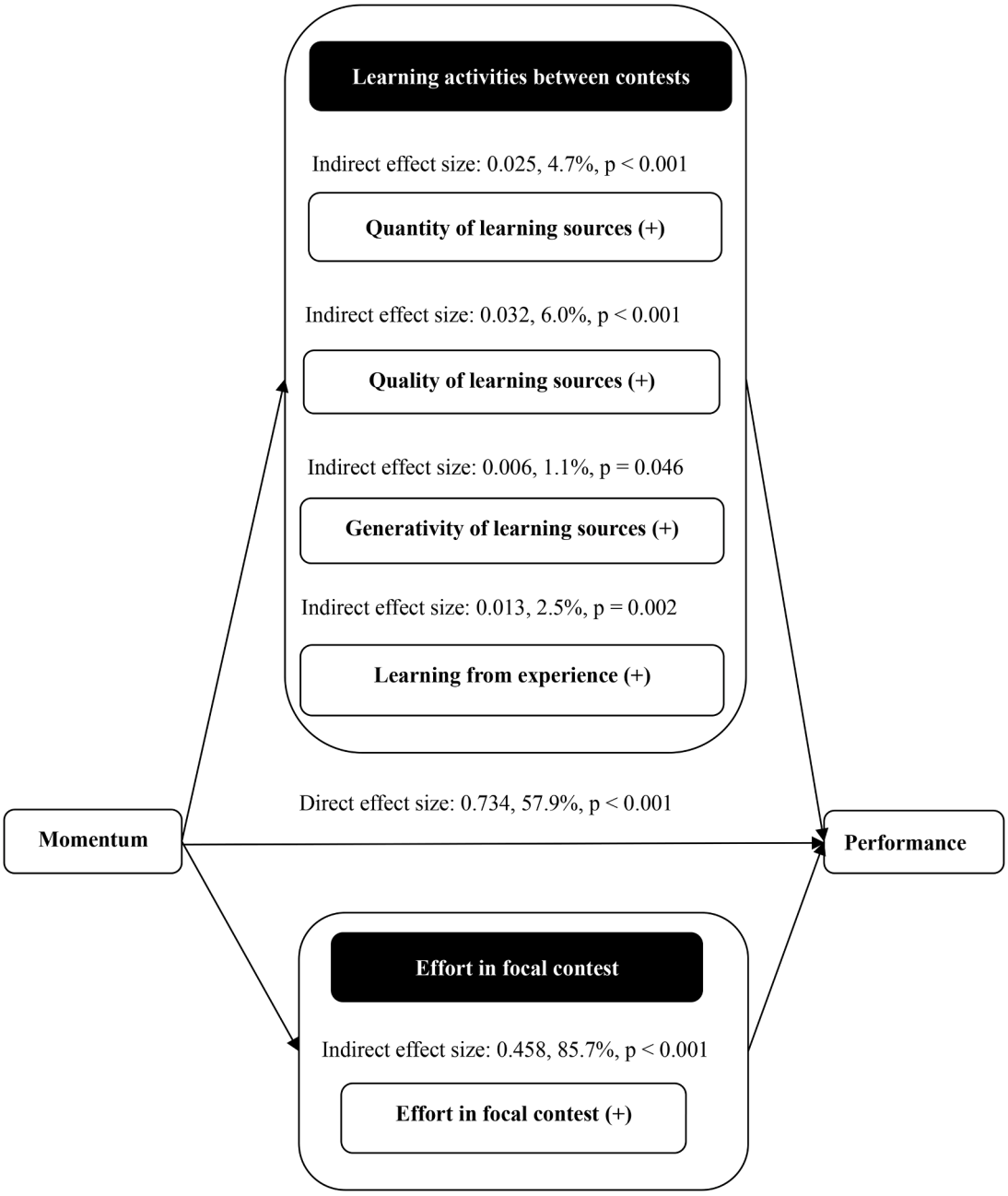


Fig. A2. Mediation analyses (momentum is measured using prior two contests).

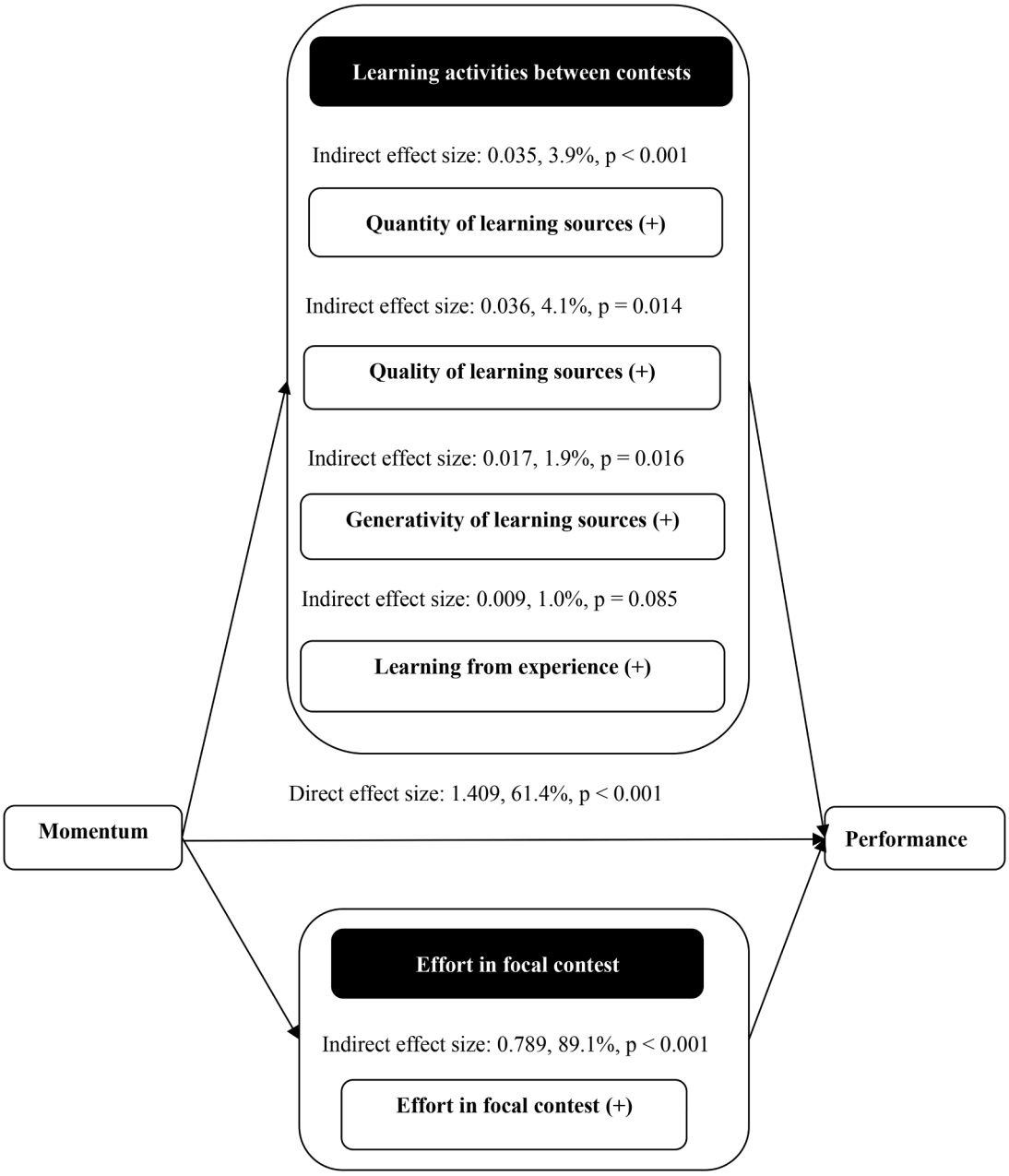


Fig. A3. Mediation analyses (momentum is measured using prior four contests).

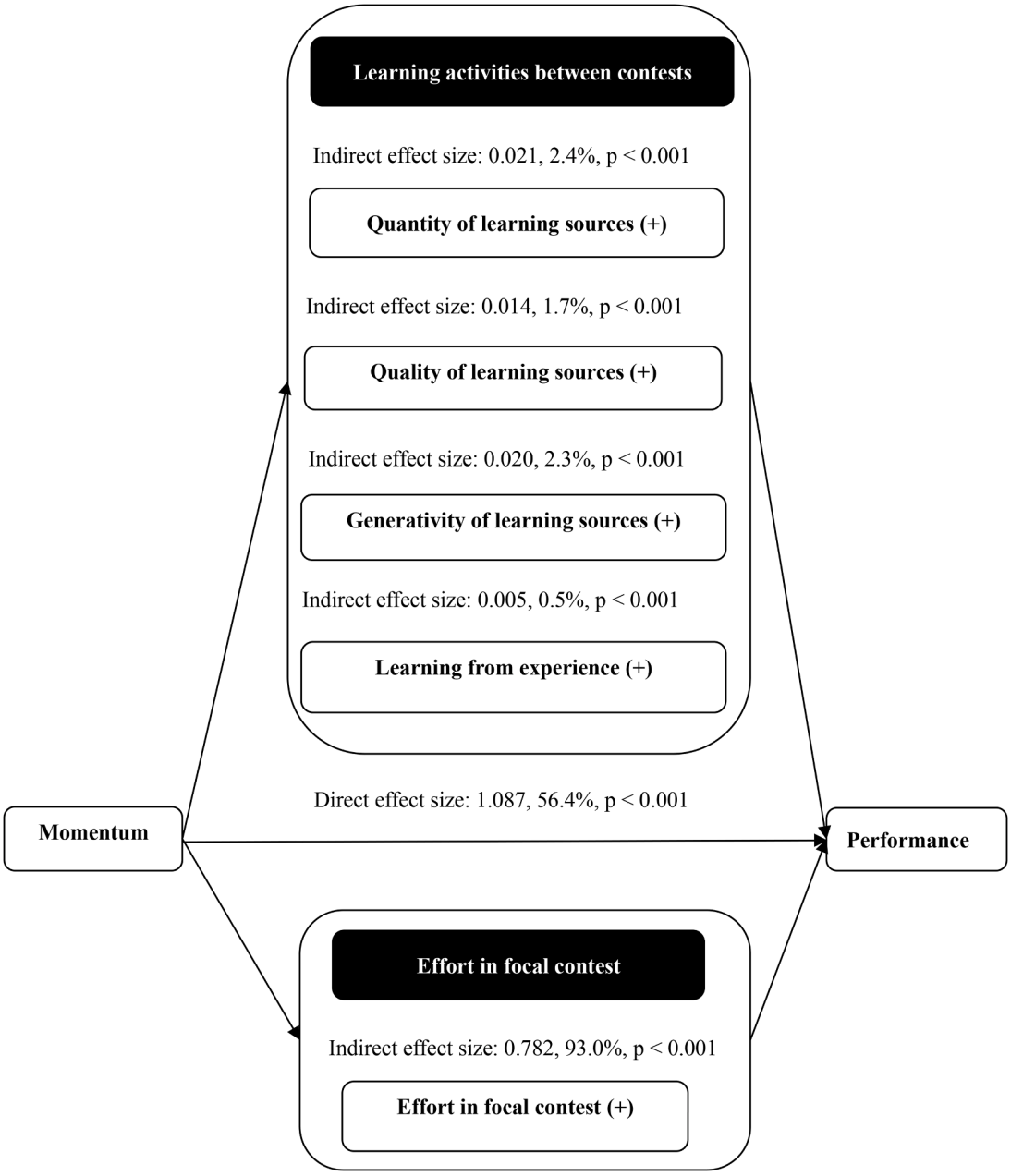


Fig. A4. Mediation analyses (performance is measured by performance scores).

References

[1] A. Ghezzi, D. Gabelloni, A. Martini, A. Natalicchio, Crowdsourcing: a review and suggestions for future research, Int. J. Manag. Rev. 20 (2018) 343–363.
[2] M. Hossain, I. Kauranen, Crowdsourcing: a comprehensive literature review, Strateg. Outsourcing: Int. J. 8 (2015) 2–22.
[3] H.C.B. Lee, S. Ba, X. Li, J. Stallaert, Salience bias in crowdsourcing contests, Inf. Syst. Res. 29 (2018) 401–418.
[4] J. Mo, S. Sarkar, S. Menon, Know when to run: recommendations in crowdsourcing contests, MIS Q. 42 (2018) 919–944.
[5] V. Grover, R.H. Chiang, T.P. Liang, D. Zhang, Creating strategic business value from big data analytics: a research framework, J. Manag. Inf. Syst. 35 (2018) 388–423.
[6] J.M. Leimeister, M. Huber, U. Bretschneider, H. Krcmar, Leveraging crowdsourcing: activation-supporting components for IT-based ideas competition, J. Manag. Inf. Syst. 26 (2009) 197–224.
[7] R. Karachiwalla, F. Pinkow, Understanding crowdsourcing projects: a review on the key design elements of a crowdsourcing initiative, Creat. Innov. Manag. 30 (2021) 563–584.
[8] Y. Zhao, Q. Zhu, Evaluation on crowdsourcing research: current status and future direction, Inf. Syst. Front. 16 (2014) 417–434.
[9] L. Chen, P. Xu, D. Liu, Effect of crowd voting on participation in crowdsourcing contests, J. Manag. Inf. Syst. 37 (2020) 510–535.
[10] P.Y. Chen, P. Pavlou, S. Wu, Y. Yang, Attracting high-quality contestants to contest in the context of crowdsourcing contest platform, Prod. Oper. Manag. 30 (2021) 1751–1771.
[11] X. Yin, K. Zhu, H. Wang, J. Zhang, W. Wang, H. Zhang, Motivating participation in crowdsourcing contests: the role of instruction-writing strategy, Inf. Manag. 59 (2022) 103616.
[12] X. Cheng, S. Fu, T. De Vreede, G.J. De Vreede, I. Seeber, R. Maier, B. Weber, Idea convergence quality in open innovation crowdsourcing: a cognitive load perspective, J. Manag. Inf. Syst. 37 (2020) 349–376.
[13] I. Dissanayake, J. Zhang, B. Gu, Task division for team success in crowdsourcing contests: resource allocation and alignment effects, J. Manag. Inf. Syst. 32 (2015) 8–39.
[14] Y. Jin, H.C.B. Lee, S. Ba, J. Stallaert, Winning by Learning? Effect of Knowledge Sharing in Crowdsourcing Contests, Inf. Syst. Res. 32 (2021) 836–859.

[15] F. Cao, W. Wang, E. Lim, X. Liu, C.W. Tan, Do social dominance-based faultlines help or hurt team performance in crowdsourcing tournaments? J. Manag. Inf. Syst. 39 (2022) 247–275.
[16] L. Crust, M. Nesti, A review of psychological momentum in sports: why qualitative research is needed, Athl. Insight 8 (2006) 1–15.
[17] S.E. Iso-Ahola, C.O. Dotson, Psychological momentum: why success breeds success, Rev. Gen. Psychol. 18 (2014) 19–33.
[18] S.E. Iso-Ahola, K.E.N. Mobily, Psychological momentum": a phenomenon and an empirical (unobtrusive) validation of its influence in a competitive sport tournament, Psychol Rep 46 (1980) 391–401.
[19] P. Adler, Momentum: A theory of Social Action, Sage, California, 1981.
[20] J. Taylor, A. Demick, A multidimensional model of momentum in sports, J Appl Sport Psychol 6 (1994) 51–70.
[21] R.J. Vallerand, P.G. Colavecchio, L.G. Pelletier, Psychological momentum and performance inferences: a preliminary test of the antecedents-consequences psychological momentum model, J. Sport Exerc. Psychol. 10 (1988) 92–108.
[22] E.H. Hwang, P.V. Singh, L. Argote, Knowl. shar. online communities: Learn. cross geogr. hierarchical boundaries Organ. Sci. 26 (2015) 1593–1611.
[23] I. Dissanayake, N. Mehta, P. Palvia, V. Taras, K. Amoako-Gyampah, Competition matters! Self-efficacy, effort, and performance in crowdsourcing teams, Inf. Manag. 56 (2019) 1–12.
[24] H. Liang, M.M. Wang, J.J. Wang, Y. Xue, How intrinsic motivation and extrinsic incentives affect task effort in crowdsourcing contests: a mediated moderation model, Comput Hum. Behav 81 (2018) 168–176.
[25] M.I. Jones, C. Harwood, Psychological momentum within competitive soccer: players' perspectives, J Appl Sport Psychol 20 (2008) 57–72.
[26] K.D. Markman, C.L. Guenther, Psychological momentum: intuitive physics and naive beliefs, Personal. Soc. Psychol. Bull. 33 (2007) 800–812.
[27] S. Miller, R. Weinberg, Perceptions of psychological momentum and their relationship to performance, Sport Psychol 5 (1991) 211–222.
[28] J. Howe, The Rise of Crowdsourcing, Wired Magazine, 2006. https://www.wired.com/2006/06/crowds/. accessed 12 January 2025.
[29] Z. Jiang, Y. Huang, D.R. Beil, The role of feedback in dynamic crowdsourcing contests: a structural empirical analysis, Manage Sci 68 (2022) 4858–4877.
[30] H. Zheng, D. Li, W. Hou, Task design, motivation, and participation in crowdsourcing contests, Int. J. Electron. Commer. 15 (2011) 57–88.
[31] R.B. Ferguson, Business Quandary? Use a Competition to Crowdsource Best Answers, MIT Sloan Manag. Rev. 54 (2013) 1.
[32] K.J. Boudreau, K.R. Lakhani, Using the crowd as an innovation partner, Harv Bus Rev 91 (2013) 60–69.
[33] J. Ye, M. Jensen, Effects of introducing an online community in a crowdsourcing contest platform, Inf. Syst. J. 32 (2022) 1203–1230.
[34] E. Segev, Crowdsourcing contests, Eur J Oper Res 281 (2020) 241–255.
[35] T.K. Koh, M.Y. Cheung, Seeker exemplars and quantitative ideation outcomes in crowdsourcing contests, Inf. Syst. Res. 33 (2022) 265–284.
[36] L. Jian, S. Yang, S. Ba, L. Lu, C. Jiang, Managing the Crowds: the Effect of Prize Guarantees and In-Process Feedback on Participation in Crowdsourcing Contests, MIS Q. 43 (2019) 97–112.
[37] S.J. Deodhar, S. Gupta, The impact of social reputation features in innovation tournaments: evidence from a natural experiment, Inf. Syst. Res. 34 (2023) 178–193.
[38] C. Patel, M.A. Husairi, C. Haon, P. Oberoi, Monetary rewards and self-selection in design crowdsourcing contests: managing participation, contribution appropriateness, and winning trade-offs, Technol Forecast Soc Change 191 (2023) 122447.
[39] I. Dissanayake, M. Yasar, M.S. Vaghefi, S.P. Nerur, Driving the Innovation Race: effect of Competitiveness in Crowdsourcing Contest, J. Manag. Inf. Syst. 41 (2024) 1142–1172.
[40] S. Wu, Q. Liu, X. Zhao, B. Sun, X. Liao, Attracting solvers' participation in crowdsourcing contests: the role of linguistic signals in task descriptions, Inf. Syst. J. 34 (2024) 6–38.
[41] J.J. Bell, C. Pescher, G.J. Tellis, J. Füller, Can AI help in ideation? A theory-based model for idea screening in crowdsourcing contests, Mark. Sci. 43 (2024) 54–72.
[42] J. Mo, N. Zhang, Framing of Seeker-Generated Information and New Solver Participation in Open Innovation Contests: an Empirical Analysis of the Temporal Effects, Inf. Syst. Res. (2025).
[43] J. Füller, K. Hutter, J. Hautz, K. Matzler, User roles and contributions in innovation-contest communities, J. Manag. Inf. Syst. 31 (2014) 273–308.
[44] K. Hutter, J. Füller, J. Hautz, V. Bilgram, K. Matzler, Machiavellianism or morality: which behavior pays off in online innovation contests? J. Manag. Inf. Syst. 32 (2015) 197–228.
[45] I. Dissanayake, J. Zhang, M. Yasar, S.P. Nerur, Strategic effort allocation in online innovation tournaments, Inf. Manag. 55 (2018) 396–406.
[46] P. Pollok, D. Lüttgens, F.T. Piller, Attracting solutions in crowdsourcing contests: the role of knowledge distance, identity disclosure, and seeker status, Res Policy 48 (2019) 98–114.
[47] S. Zhang, P.V. Singh, A. Ghose, A structural analysis of the role of superstars in crowdsourcing contests, Inf. Syst. Res. 30 (2019) 15–33.
[48] S.J. Deodh, Y. Babar, G. Burtch, The influence of status on evaluations: evidence from online coding contests, MIS Q. 46 (2022) 2085–2110.
[49] B. Yan, A.B. Hollingshead, Motivating the Motivationally Diverse Crowd: social Value Orientation and Reward Structure in Crowd Idea Generation, J. Manag. Inf. Syst. 39 (2022) 1064–1088.
[50] P. Kumar, S.J. Deodhar, S. Zaheer, Cognitive sources of liability of foreignness in crowdsourcing creative work, J Int Bus Stud 54 (2023) 686–716.
[51] H.C.B. Lee, A. Mishra, Lost time in crowdsourcing contests, Inf. Syst. Res. (2025).
[52] N.C. Pettit, N. Sivanathan, E. Gladstone, J.C. Marr, Rising stars and sinking ships: consequences of status momentum, Psychol Sci 24 (2013) 1579–1584.
[53] H. Kakkar, N. Sivanathan, N.C. Pettit, The impact of dynamic status changes within competitive rank-ordered hierarchies, Proc. Natl. Acad. Sci. 116 (2019) 23011–23020.
[54] B.L. Bayus, Crowdsourcing new product ideas over time: an analysis of the Dell IdeaStorm community, Manage Sci 59 (2013) 226–244.
[55] M. Li, A. Kankanhalli, S.H. Kim, Which ideas are more likely to be implemented in online user innovation communities? an empirical analysis, Decis Support Syst 84 (2016) 28–40.
[56] Q. Liu, Q. Du, Y. Hong, W. Fan, S. Wu, User idea implementation in open innovation communities: evidence from a new product development crowdsourcing community, Inf. Syst. J. 30 (2020) 899–927.
[57] T.L. Hubbard, Toward a general theory of momentum-like effects, Behav, Process. 141 (2017) 50–66.
[58] I.Y. Nahm, M.J. Ahearne, N. Lee, S. Tirunillai, Managing Positive and Negative Trends in Sales Call Outcomes: the Role of Momentum, J. Mark. Res. 59 (2022) 1120–1140.
[59] Y. Zhao, H. Zhang, Does success breed success? An investigation of momentum in elite recurve archery, Psychol Sport Exerc 66 (2023) 102397.
[60] C. Gernigon, W. Briki, K. Eykens, The dynamics of psychological momentum in sport: the role of ongoing history of performance patterns, J. Sport Exerc. Psychol. 32 (2010) 377–400.
[61] D.W. Lehman, J. Hahn, Momentum and organizational risk taking: evidence from the National Football League, Manage Sci 59 (2013) 852–868.
[62] G.P. Latham, C.C. Pinder, Work motivation theory and research at the dawn of the twenty-first century, Annu Rev Psychol 56 (2005) 485–516.
[63] M.V. Covington, Goal theory, motivation, and school achievement: an integrative review, Annu Rev Psychol 51 (2000) 171–200.
[64] R.J. Vallerand, Intrinsic and extrinsic motivation in sport and physical activity: a review and a look at the future. Handbook of Sport Psychology, 2007, pp. 59–83.
[65] R. Kanfer, M. Frese, R.E. Johnson, Motivation related to work: a century of progress, J. Appl. Psychol. 102 (2017) 338–355.
[66] J.A. Bargh, Losing consciousness: automatic influences on consumer judgment, behavior, and motivation, J. Consum. Res. 29 (2022) 280–285.
[67] D.A. Kolb, Experiential learning: Experience as the Source of Learning and Development, FT press, 2014.
[68] I. Blohm, J.M. Leimeister, H. Krcmar, Crowdsourcing: how to benefit from (too) many great ideas, MIS Q. Exec. 12 (2013) 199–211.
[69] T.L. Hubbard, The varieties of momentum-like experience, Psychol Bull 141 (2015) 1081–1119.
[70] J.M. Shaw, D.A. Dzewaltowski, M. McElroy, Self-Efficacy and Causal Attributions as Mediators of Perceptions of Psychological Momentum, J Sport Exerc Psychol 14 (1992) 134–147.
[71] E.D. Deemer, P.A. Derosa, S.A. Duhon, A.M. Dotterer, Psychological momentum and inertia: toward a model of academic motivation, J Career Dev 48 (2021) 275–289.
[72] D. Miller, J. Shamsie, Learning across the life cycle: experimentation and performance among the Hollywood studio heads, Strateg. Manag. J. 22 (2001) 725–745.
[73] D. Boud, H. Middleton, Learning from others at work: communities of practice and informal learning, J. Workplace Learn. 15 (2003) 194–202.
[74] R. Reagans, L. Argote, D. Brooks, Individual experience and experience working together: predicting learning rates from knowing who knows what and knowing how to work together, Manage Sci 51 (2005) 869–881.
[75] C.C. Manz, H.P. Sims Jr, Vicarious learning: the influence of modeling on organizational behaviour, Acad. Manag. Rev. 6 (1981) 105–113.
[76] P.V. Mannucci, K. Yong, The differential impact of knowledge depth and knowledge breadth on creativity over individual careers, Acad. Manag. J. 61 (2018) 1741–1763.
[77] D. Kyoon Yoo, M.A. Vonderembse, T.S. Ragu-Nathan, Knowledge quality: antecedents and consequence in project teams, Journal of Knowledge Management 15 (2011) 329–343.
[78] P. Ritala, C. De Kort, B. Gailly, Orchestrating knowledge networks: alter-oriented brokering, J Manage 49 (2023) 1140–1178.
[79] D.A. Kolb, Experiential learning: Experience as the Source of Learning and Development, Prentice Hall, New Jersey, 1984.
[80] D.A. Kolb, R.E. Boyatzis, C. Mainemelis, Experiential learning theory: previous research and new directions, in: R.J. Sternberg, L.F. Zheng (Eds.), Perspectives On Thinking, Learning, and Cognitive Styles, Routledge, 2014, pp. 227–248.
[81] A.F. Hayes, Introduction to mediation, moderation, and Conditional Process Analysis a Regression-Based Approach, The Guilford Press, New York, 2013.
[82] Z. Lei, D. Yin, H. Zhang, Focus within or on others: the impact of reviewers' attentional focus on review helpfulness, Inf. Syst. Res. 32 (2021) 801–819.
[83] K. Lim, H. Kwon, The effects of emotional leadership on innovative behavior and psychological depression of bankers: analysis of double mediation effect, J. Korea Ind. Inf. Syst. Res. 24 (2019) 107–118.
[84] M. Sosna, R.N. Trevinyo-Rodríguez, S.R. Velamuri, Business model innovation through trial-and-error learning: the Naturhouse case, Long Range Plann 43 (2010) 383–407.
[85] X. Zheng, G. Zhou, D. Zeng, Platform governance in the era of AI and the digital economy, Front. Eng. Manag. 10 (2023) 177–182.
[86] M. Wang, Z. Zhang, Emergence mechanisms of group consensus in social networks, Front. Eng. Manag. 11 (2024) 362–368.

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